```python
import numpy as np
import random
import matplotlib.pyplot as plt
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Dense, Flatten
```

Load Dataset

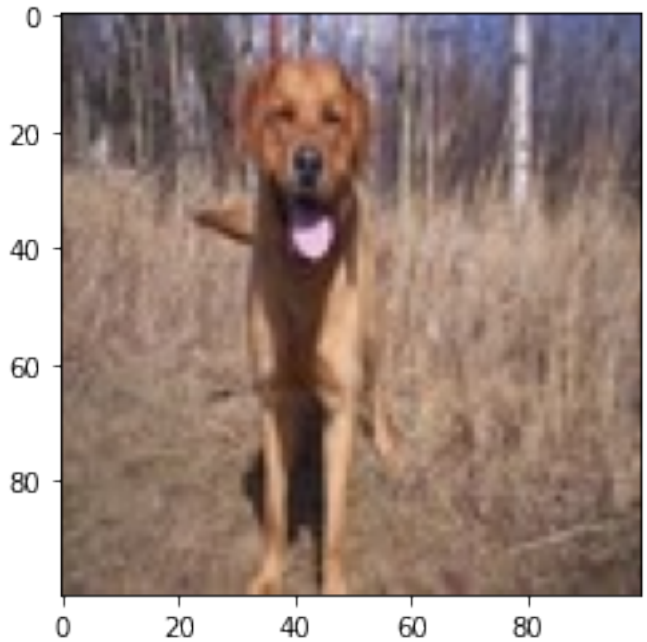
```python
X_train = np.loadtxt('input.csv', delimiter = ',')
Y_train = np.loadtxt('labels.csv', delimiter = ',')

X_test = np.loadtxt('input_test.csv', delimiter = ',')
Y_test = np.loadtxt('labels_test.csv', delimiter = ',')
```

```python
X_train = X_train.reshape(len(X_train), 100, 100, 3)
Y_train = Y_train.reshape(len(Y_train), 1)

X_test = X_test.reshape(len(X_test), 100, 100, 3)
Y_test = Y_test.reshape(len(Y_test), 1)

X_train = X_train/255.0
X_test = X_test/255.0
```

```python
print("Shape of X_train: ", X_train.shape)
print("Shape of Y_train: ", Y_train.shape)
print("Shape of X_test: ", X_test.shape)
print("Shape of Y_test: ", Y_test.shape)
```

```
Shape of X_train:  (2000, 100, 100, 3)
Shape of Y_train:  (2000, 1)
Shape of X_test:  (400, 100, 100, 3)
Shape of Y_test:  (400, 1)
```

```python
idx = random.randint(0, len(X_train))
plt.imshow(X_train[idx, :])
plt.show()
```



Model

```
model = Sequential([
    Conv2D(32, (3,3), activation = 'relu', input_shape = (100, 100, 3)),
    MaxPooling2D((2,2)),

    Conv2D(32, (3,3), activation = 'relu'),
    MaxPooling2D((2,2)),

    Flatten(),
    Dense(64, activation = 'relu'),
    Dense(1, activation = 'sigmoid')
])
```

```
model = Sequential()

model.add(Conv2D(32, (3,3), activation = 'relu', input_shape = (100, 100, 3)))
model.add(MaxPooling2D((2,2)))

model.add(Conv2D(32, (3,3), activation = 'relu'))
model.add(MaxPooling2D((2,2)))

model.add(Flatten())
```

```
model.add(Dense(64, activation = 'relu'))
model.add(Dense(1, activation = 'sigmoid'))
```

```
model.compile(loss = 'binary_crossentropy', optimizer = 'adam', metrics = ['accuracy'])
```

```
model.fit(X_train, Y_train, epochs = 5, batch_size = 64)
```

```
Epoch 1/5
32/32 [==============================] - 30s 934ms/step - loss: 0.3599 - accuracy: 0.8475
Epoch 2/5
32/32 [==============================] - 29s 911ms/step - loss: 0.2853 - accuracy: 0.8800
Epoch 3/5
32/32 [==============================] - 30s 946ms/step - loss: 0.2196 - accuracy: 0.9160
Epoch 4/5
32/32 [==============================] - 32s 994ms/step - loss: 0.1694 - accuracy: 0.9395
Epoch 5/5
32/32 [==============================] - 30s 931ms/step - loss: 0.1247 - accuracy: 0.9610
```

```
<keras.callbacks.History at 0x1de37b90850>
```

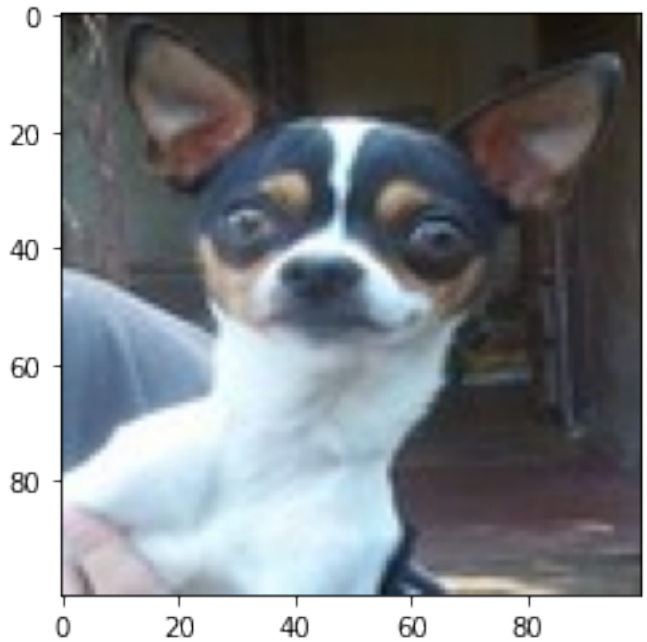
```
model.evaluate(X_test, Y_test)
```

```
13/13 [==============================] - 4s 149ms/step - loss: 0.9762 - accuracy: 0.6550
```

```
[0.9762333631515503, 0.6549999713897705]
```

Making predictions

```
idx2 = random.randint(0, len(Y_test))
plt.imshow(X_test[idx2, :])
plt.show()

y_pred = model.predict(X_test[idx2, :].reshape(1, 100, 100, 3))
y_pred = y_pred > 0.5

if(y_pred == 0):
    pred = 'dog'
else:
    pred = 'cat'

print("Our model says it is a :", pred)
```




```
Our model says it is a : dog
```